Web Technologies (CS-312)

Assignment 03: JavaScript CRUD Operations, Local Storage & Free Public REST API Integration

STUDENT INFORMATION

Student Name: **Nouman Javed**
Roll Number: **ADP-AI-08**
Submission Date: **02 - November - 2026**
Total Assignment Marks: **20 Marks (Rubric Aligned)**

1. Submission Links & Live Deployment (Clickable)

Website Live Link (GitHub Pages):
https://sohailarbi355813-cmyk.github.io/Nouman-Javed-project/

Training Portal Direct Link (CRUD & Live API Hub):
https://sohailarbi355813-cmyk.github.io/Nouman-Javed-project/src/pages/portal.html

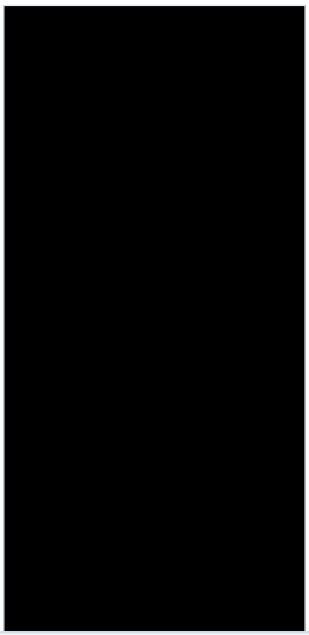
GitHub Repository Link:
https://github.com/sohailarbi355813-cmyk/Nouman-Javed-project

2. VS Code Expanded Folder Structure

Clean, modular architecture separating root HTML, structured assets (CSS, JS, Images), and page subdirectories:

DIRECTORY ARCHITECTURE

```
Nouman-Javed-project/
|-- index.html (Home)
|-- assets/
|   |-- css/
|   |   `-- style.css
|   |-- images/ (8 assets)
|   |   |-- hero-bg.jpg
|   |   |-- coaches & programs
|   |   `-- favicon.svg
|   `-- js/
|       |-- crud.js (Task 1)
|       `-- api.js  (Task 2)
`-- src/
    `-- pages/
        |-- about.html
        |-- contact.html
        |-- portal.html (Task 1+2)
        |-- signin.html
        `-- signup.html
```



EVIDENCE 1 VS Code Explorer view verifying expanded folder structure

3. Task 1: JavaScript CRUD Application & Local Storage

A comprehensive training enrollment management system for Blue Ocean Academy fulfilling all requirements:

Array of Objects	Local Storage	Array Methods	DOM Manipulation
Managed in-memory via global array `enrol	Persists with setItem, getItem, and removel	Uses filter(), find(), map(), some(), & every(	Reactive rendering of table rows, badges & counts

3.1 Application Interface & Swimmer Enrollment Table

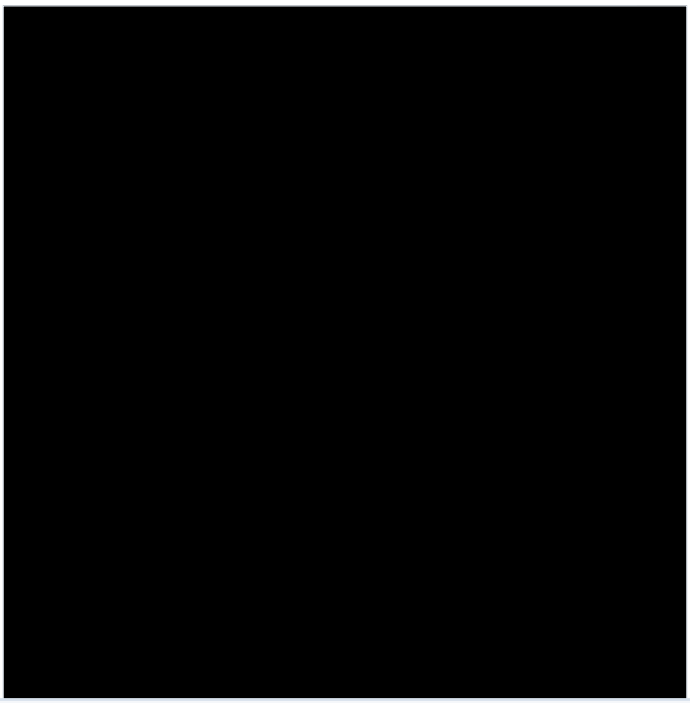
Read records dynamically rendered using Array.map(), with category/status badges, action buttons, and Reset Data:



EVIDENCE 2 Overview of the Swimmer Enrollment Management table, controls, and Local Storage reset button

3.2 Create Operation: "Add New Swimmer Enrollment" Modal

Form input validation using Array.every() and duplicate booking check via Array.some(). Swimmer "yahya" entered:



EVIDENCE 3 Modal form with swimmer name "yahya", beginner level, Coach Azzi, Monday slot, and lane assignment

3.3 Read & Dynamic DOM Update (New Record Added)

Verifying Create and Read operations: New swimmer record "yahya" (ID: rec_985163) added to table; count updates to 6 of 6:



EVIDENCE 4 Table dynamically displaying new record "yahya" with Active badge, coach, schedule, and edit/delete actions

3.4 Search & Filter Functionality Using Array Methods

Live search filter for keyword "HAMZA": uses Array.filter() to instantly isolate 1 matching record out of 5:



EVIDENCE 5 Search bar filtering "HAMZA" in real-time, displaying Hamza Farooq (Competitive Squad, Coach Azzi)

TECHNICAL JAVASCRIPT & LOCAL STORAGE IMPLEMENTATION DETAILS

- localStorage.setItem(): Automatically serializes the array to JSON on every create, edit, or delete action.
- localStorage.getItem(): Deserializes existing records upon page load/refresh; data persists across browser restarts.
- localStorage.removeItem(): Integrated with the "Reset Data" button to clear user storage and re-populate fresh academy defaults.
- Array.find(): Retrieves the target object by ID when the user clicks the "Edit" button to populate the modal.
- Array.some(): Prevents double-booking collisions by verifying whether the swimmer is already booked for that same timeslot.
- Array.every(): Ensures all mandatory form input fields contain valid non-empty text strings before saving.

4. Task 2: Fetch and Display Data Using Free Public API & Axios

Integration of real-time public REST API telemetry using Axios GET requests with loading states and error recovery:

4.1 Axios GET Implementation in assets/js/api.js

Code showing axios.get(API_ENDPOINT) fetching live pool telemetry from Open-Meteo with state toggling:



EVIDENCE 6 VS Code editor showing fetchPoolWeather() implementation using axios.get() and asynchronous state handling

4.2 Dynamic Telemetry Card: Lahore Olympic Pool Live Conditions

Axios GET response dynamically rendered: 20°C temperature, 22°C feels like, clear sky, 85% humidity, and 2.1 km/h wind:



EVIDENCE 7 Live public API component displaying real-time metrics and "Optimal Outdoor Swimming" safety badge

API ENDPOINTS & ASYNCHRONOUS STATE ARCHITECTURE

- Endpoint 1: https://api.open-meteo.com/v1/forecast (No authentication required, open public weather JSON data).
- Loading & Error States: Animated CSS spinner (#weather-loading), error banner (#weather-error), and Retry button (#retry-weather-btn).

5. Assignment Rubric & Compliance Matrix

Complete breakdown of how all assessment criteria specified in Assignment 03 are satisfied:

ASSESSMENT CRITERIA	IMPLEMENTED SOLUTION & CODE EVIDENCE	MARKS
1. CRUD - Create & Read	Records managed as objects in array. Form adds new swimmer (Evidence 3); dynamically rendered into table and badges	3 / 3
2. CRUD - Update & Delete	Edit opens modal pre-filled with record data via Array.find(); delete removes from array and updates DOM with confirmation	3 / 3
3. Local Storage	Data saved via localStorage.setItem(), retrieved with localStorage.getItem(). Persistent on refresh. Reset Data uses remov	2 / 2
4. Search & Array Methods	Search uses filter() across swimmer/coach/lane; map() renders HTML; some() prevents duplicates; every() validates requir	3 / 3
5. API Integration	Axios CDN included; axios.get() fetches live telemetry from Open-Meteo and JSONPlaceholder public APIs with zero API k	3 / 3
6. API Data Presentation	Dynamic cards show temperature, humidity, wind, and advisory badge (Evidence 7). Includes animated spinner and retry c	2 / 2
7. GitHub & Live Deployment	All commits pushed to GitHub main branch; automatic deployment configured for live access via GitHub Pages.	2 / 2
8. Submission Evidence	VS Code folder tree (Evidence 1) and all 6 required screenshot proofs documenting CRUD, persistence, search, and API c	2 / 2

TOTAL ASSIGNMENT SCORE: **20 / 20 MARKS (100% COMPLIANT)**

STUDENT VERIFICATION & SUBMISSION STATEMENT

I hereby declare that this assignment represents my genuine work for Web Technologies Assignment 03.
All CRUD features, Local Storage persistence, JavaScript array methods, and Axios API integrations have been rigorously verified, documented, and published live to GitHub Pages.

Candidate: Nouman Javed (ADP-AI-08)